Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/16/2019
PI Name:	Smith, Scott M Ph.D.		
Project Title:	Integrated Nutrition: Development and Provision of Functional Foods to Mitigate Bone Loss on ISS Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/13/2013	**End Date:**	12/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2018 per PI (Ed., 11/4/19) NOTE: Risk/Gap changes per IRP Rev E (Ed., 3/24/14)		
Key Personnel Changes/Previous PI:	September 2017 report: Maya Cooper and Takiyah Sirmons are added as CoInvestigators.		
COI Name (Institution):	Zwart, Sara Ph.D. (Universities Space Research Association/NASA JSC) Douglas, Grace Ph.D. (NASA Johnson Space Center) Heer, Martina Ph.D. (RFW-University of Bonn) Kloeris, Vickie M.S. (NASA Johnson Space Center) Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association/NASA JSC) Sirmons, Takiyah Ph.D. (Leidos/NASA Johnson Space Cente) Cooper, Maya M.S. (Leidos/NASA Johnson Space Cente)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The role of nutrition in human health, on Earth or in space, is incontrovertible. Although space flight and ground-based studies have shown the role of individual nutrients, and even specific food types, on human systems, a broad study of dietary intake in relation to human adaptation to space flight has not been conducted. We propose here a multifaceted effort to conduct a comprehensive assessment of dietary intake during long duration International Space Station (ISS) missions, coupled with an effort to develop health-promoting, highly acceptable, functional food options that do not exist in the nominal space food system. We will work with crewmembers to guide them on how to improve their nutritional intake. Optimized dietary intake has the potential to mitigate risks to virtually every physiological system and other risks associated with space flight. Nutritional benefits are expected for bone, muscle, cardiovascular, immune, renal, and gastrointestinal systems. Additionally, optimizing dietary intake is expected to help mitigate oxidative damage associated with space flight, along with radiation risks. Beyond physiology and biochemistry, healthier diets and improved overall intake are expected to enhance crew morale, performance, and even sleep. This project has 3 elements: 1) developing health-promoting functional foods for space flight, 2) educating crewmembers on the importance of nutrition and functional foods in health and adaptation to space flight, and providing such foods for their ISS missions, and 3) assessing detailed dietary intake over the course of ISS flight in relation to health outcomes. Because prescribing precise diets for a 6-month mission is not realistic, we will work with crews to target healthy/functional food provision to meet their individual preferences, and to provide guidance on how to best incorporate these items into their menu selections. This feedback will be provided in planning for their mission, as well as in near real-time during their mission. We aim to provide optimal nutrition with foods rich in beneficial nutrients, as opposed to supplements. This objective is based on ground-based evidence that studies using supplements time and again fail to produce the same results as studies (controlled or epidemiological) using whole foods. On orbit, intake data will be recorded and downlinked, and periodic feedback will be provided to the crew with additional suggestions for improvement. Intake data from the mission will ultimately be evaluated and a number of nutrients will be assessed and the results compared with health outcome data. For this project, we will focus on bone as the primary outcome factor, as it is assumed that other projects will be solicited to study the relationship of diet on other systems and crew performance outcomes. This study will be the first comprehensive test evaluating the extent to which nutrition may mitigate the negative effects of space flight, and will provide the background for developing food and nutrition plans for future exploration missions.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Osteoporosis is a global health concern. Testing an optimal diet in an environment with accelerated bone loss has advantages in that we expect to observe dietary effects faster than what normally could be observed on Earth. This study will provide a comprehensive assessment of dietary intake during ISS missions, coupled with an effort to provide healthier food options and to guide crewmembers on how to improve their dietary intake, to mitigate risks to virtually every physiological system, with a focus on bone. Results from this study will not only benefit astronauts on long-duration space missions, but will also benefit people on Earth.
Task Progress:	Project complete. New functional space foods have been developed. The flight evaluation of effectiveness was cancelled years ago. The execution of the study involved a shelf life assessment of potential functional space foods. The stability of lycopene, flavonoids, and omega-3 fatty acids were assessed through two-years of storage (21°C) in 5 space food items currently processed by the Space Food Systems Laboratory (SFSL), 10 selected commercial products, and 18 recently developed functional foods. Samples were sent to the Food Composition Laboratory of the Linus Pauling Institute at Oregon State University (Corvallis, OR) for bioactive compound analysis. Sensory evaluation was conducted with untrained panelists (n = 25) to assess the organoleptic properties of each product. Duplicate samples of the food were placed in 21°C environmental chambers within the Space Food Systems Laboratory for controlled storage and re-analysis of nutritional and sensory properties after 1-year and 2-year periods. Functional space foods to support bone health are feasible for development given the stability of lycopene, phenolic compounds, and omega-3 fatty acid concentrations in space foods after 2 years of storage at 21°C. The stability of anthocyanins and vitamin K may vary with food product. There was no evidence from the study suggesting that thermostabilized, freeze-dried, or natural form foods should have predominance over the other food forms due to bioactive compound stability. Additionally, the developed food products of this study demonstrated successful expansion of the fruit, vegetable, and seafood offerings of the space food system. Furthermore, new functional foods exceeded minimum acceptability standards for sensory attributes such as color, aroma, taste, texture, appearance, and overall acceptability. Organoleptic acceptability was sustained in most of the functional foods in this study for 2 years. Next steps include establishing bioactive compound target intakes for efficacy such that appropriate foods and variety can be added to the space food system.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
NASA Technical Documents	Smith SM, Zwart SR, Douglas GL, Heer M. "Human adaptation to spaceflight: The role of food and nutrition. Second edition." Houston, TX: NASA Lyndon B. Johnson Space Center, 2021. 255 p. NP-2021-03-003-JSC. https://www.nasa.gov/sites/default/files/atoms/files/human_adaptation_2021_final.pdf , Apr-2021